

Deftaudio Ammeter

User Manual 1.0

Overview

The Ammeter is a Eurorack utility module designed to monitor real-time current consumption across your system's power rails. It helps identify power-related issues such as instability, noise, and overload conditions by providing accurate, visual feedback.

Unlike relying on manufacturer specifications (which often list average current draw), Ammeter reveals dynamic power behavior which is critical for modern systems with digital or variable-load modules.

Key Features

- Dual analog needle ammeters for real-time current monitoring
- Monitors:
 - +12V rail (fixed, always active)
 - -12V or +5V rail (user selectable)
- Functions as a bus board (busbar) for distributing power
- Replaceable fuses per rail for protection
- Various current range per meter (500mA to 3A) at purchase
- Visual, high-resolution feedback via analog needles

Understanding Eurorack Power

- A typical Eurorack system uses three voltage rails:
 - +12V – primary rail for analog and many digital modules
 - -12V – used in analog circuitry
 - +5V – mostly used by digital/computer-based modules

Manufacturers usually specify average current draw, but real-world usage can vary significantly depending on:

- CPU activity (digital modules)
- Signal processing intensity

Ammeter allows you to observe these fluctuations in real time.

Installation

Mounting

- Install the Ammeter module into your Eurorack case like any standard module.

Power Connection

- Connect your power supply to the input connector (**lowest connector**) on the Ammeter using a standard Eurorack ribbon cable.
- Connect your modules to the Ammeter's output headers.
- (Optional) Cascade additional bus boards if more connectors are needed.
- The Ammeter must sit between the power source and modules to measure current correctly.

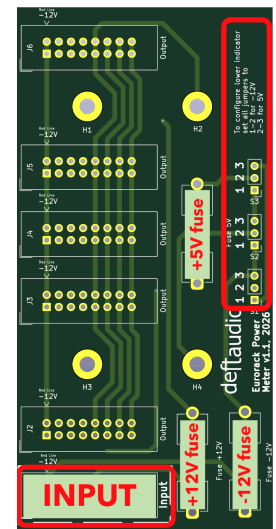
Configuration

Configuring lower meter (+5V or -12V Selection)

- On the back of the module set jumpers to select whether the lower meter monitors:
 - -12V, or
 - +5V

Fuse Settings

- Each rail includes a replaceable fuse:
- Set fuse ratings according to your system limits
- Replace blown fuses as needed



Safety Notes

Always power down your system before connecting or reconfiguring

Ensure correct ribbon cable orientation

Use appropriate fuse ratings

Summary

The Ammeter provides a simple but powerful way to see what your power system is actually doing, not just what specifications suggest. Its real-time analog display makes it especially effective for catching transient behavior that could otherwise go unnoticed.